Lab 7: Extending OLS in Social Science Research

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Instructions

This lab has two parts. In **Part 1**, you'll estimate panel data models — pooled OLS, fixed effects, and random effects — and learn how to choose among them. In **Part 2**, you'll explore two strategies for establishing causal relationships: mediation analysis and instrumental variables.

Work through each task in order. Code scaffolding is provided — fill in the `...` placeholders with the correct code. After running each model, take a moment to interpret the output before moving on. Written reflection questions appear at the end.

Setup

Start by setting your working directory, clearing the workspace, and loading the packages you'll need. If you haven't installed them before, uncomment the `install.packages()` line and run it once.

```
## Set your working directory to the ICPSR_regression folder
## Change the path below to match your computer
# setwd("~/ICPSR_regression")
setwd("C:/Users/Patrick Shea/Dropbox/My Projects/Spring 2026/ICPSR/ICPSR_OLS26")

rm(list = ls())

# install.packages(c("plm", "haven", "stargazer", "mediation", "AER"))
library(plm)
library(dplyr)
library(haven)
library(stargazer)
```

Part 1: Panel Data Analysis

You'll analyze a dataset on **charitable giving** that tracks 47 individuals from 1979 to 1988. Because the same people are observed repeatedly, this is **panel data** — and ignoring that structure can lead to biased estimates.

Variables

Variable	Description
`charity`	Sum of cash and property contributions
`income`	Adjusted gross income
`price`	One minus the marginal tax rate
`age`	1 if taxpayer is over 64, 0 otherwise
`ms`	1 if married, 0 otherwise
`deps`	Number of dependents

Task 1.1: Load and examine the data

Load the dataset and look at its structure. Then tell R this is panel data using `pdata.frame()`.

```
charity <- read_dta("data/charity.dta")

# Examine the data
head(charity)
str(charity)
```

Hint

The panel identifier variables are `subject` (who) and `time` (when). Use `pdata.frame()` to declare the panel structure:

```
charity <- pdata.frame(charity, index = c("subject", "time"))
```

After declaring the panel structure, run `summary(charity)` to see what you're working with.

Question: What makes this a panel dataset? How many individuals are there, and over how many time periods?

Task 1.2: Pooled OLS

Estimate a standard OLS model that ignores the panel structure entirely. This treats every person-year observation as independent.

```
pooled_model <- lm(... , data = charity)
summary(pooled_model)
```

Hint

The formula is: `charity ~ age + income + price + deps + ms`. This is just regular `lm()` — the same function you've used all course.

Think about what this model assumes. It treats a wealthy person in 1985 and a low-income person in 1980 as if they're drawn from the same population with no individual-level differences. Is that realistic?

Task 1.3: Fixed effects

Now estimate a **fixed effects** model using the `plm()` function. Fixed effects controls for all time-invariant characteristics of each individual by comparing each person *to themselves* over time.

```
fe_model <- plm(... , data = charity, model = "within")
summary(fe_model)
```

Hint

The formula is the same as pooled OLS: `charity ~ age + income + price + deps + ms`. The key difference is specifying `model = "within"` in `plm()`. If you already declared the panel structure with `pdata.frame()`, you don't need to specify `index` again.

Compare the output to pooled OLS. Some things to notice:

- Is there an intercept? (Why or why not?)
- Did any variables disappear or change dramatically?
- How do the coefficient magnitudes and significance levels compare?

Task 1.4: Random effects

Estimate a **random effects** model. This assumes that individual-specific effects are random draws from a distribution and — critically — are uncorrelated with the predictors.

```
re_model <- plm(... , data = charity, model = "random")
summary(re_model)
```

Hint

Same formula again — just change `model = "within"` to `model = "random"`.

Task 1.5: Choose your model

Now use the **Hausman test** to decide between fixed and random effects.

- **H0:** Individual effects are uncorrelated with the predictors (random effects is consistent).
- **H1:** Correlation exists (only fixed effects is consistent).

```
phtest(fe_model, re_model)
```

Hint

The decision rule is straightforward: if $p < 0.05$, reject H0 and use fixed effects. If $p > 0.05$, random effects is acceptable.

You can also run a Breusch-Pagan test to check whether you need panel models at all (versus plain OLS):

```
plmtest(re_model, type = "bp")
```

Question: Which model should you use and why? How do the coefficients differ across the three models?

Bonus: Compare all three models

Put the three models side-by-side in a regression table to see the differences clearly:

```
stargazer(pooled_model, fe_model, re_model, type = "text",
          column.labels = c("Pooled OLS", "Fixed Effects", "Random Effects"),
          title = "Comparison of Panel Data Models")
```

Part 2: Two-Stage Models

You'll use the dataset from Miguel, Satyanath, and Sergenti (2004), which studies how economic shocks driven by **rainfall** affect **civil conflict** in Africa. This is a landmark paper in the causal inference literature.

Variables

Variable	Description
`any_prio`	Civil conflict indicator (0 = no, 1 = yes)
`gdp_g`	GDP growth rate
`GPCP_g`	Rainfall growth (current year)
`GPCP_g_l`	Rainfall growth (previous year)
`polity2l`	Democracy level
`ethfrac`	Ethnic fractionalization
`Oil`	Oil rents
`lpopl1`	Log population

```
library(mediation)
library(AER)

conflict_data <- read_dta("data/rainconflict.dta")
head(conflict_data)
summary(conflict_data[, c("any_prio", "gdp_g", "GPCP_g", "polity2l",
                          "ethfrac", "Oil", "lpopl1")])
```

Task 2.1: Mediation analysis

Research question: *Does rainfall affect conflict through its impact on economic growth?*

The hypothesized causal chain is:

$$\text{Rainfall} \longrightarrow \text{GDP Growth} \longrightarrow \text{Conflict}$$

But rainfall might also affect conflict directly (not through growth). Mediation analysis separates these two pathways.

- **Treatment (X):** `GPCP_g` (rainfall)
- **Mediator (M):** `gdp_g` (GDP growth)
- **Outcome (Y):** `any_prio` (conflict)
- **Controls:** `polity2l + ethfrac + Oil + lpopl1`

Step 1: Estimate the **mediator model** — does rainfall predict GDP growth?

```
med_model <- lm(gdp_g ~ ... , data = conflict_data)
summary(med_model)
```

> Hint
>
> The mediator model regresses the mediator (`gdp_g`) on the treatment (`GPCP_g`) plus all controls: `gdp_g ~ GPCP_g + polity2l + ethfrac + Oil + lpopl1`.

Step 2: Estimate the **outcome model** — do both rainfall and growth predict conflict?

```
out_model <- lm(any_prio ~ ... , data = conflict_data)
```

```
out_model <- lm(any_prio ~ ... , data = conflict_data)
summary(out_model)
```

> **Hint**
>
> The outcome model includes *both* the treatment and the mediator on the right-hand side: `any_prio ~ GPCP_g + gdp_g + polity2l + ethfrac + Oil + lpopl1`.

Step 3: Run the formal **mediation analysis** using the `mediate()` function. This uses bootstrapping to compute confidence intervals for the indirect, direct, and total effects.

```
result <- mediate(med_model, out_model,
                  treat = "GPCP_g", mediator = "gdp_g",
                  sims = 1000, boot = TRUE)
summary(result)
plot(result)
```

> **Reading the output**
>
> The `mediate()` output reports four quantities:
>
> - **ACME** (Average Causal Mediation Effect) = the indirect effect through GDP growth
> - **ADE** (Average Direct Effect) = the effect of rainfall on conflict that does *not* pass through growth
> - **Total Effect** = ACME + ADE
> - **Proportion Mediated** = ACME / Total Effect
>
> Check whether each effect is statistically significant (look at the confidence intervals and p-values).

Questions:

- What is the direct effect of rainfall on conflict?
- What is the indirect effect through GDP growth?
- Which pathway is stronger?

Task 2.2: Instrumental variables

Research question: *What is the causal effect of economic growth on conflict?*

The problem: GDP growth and conflict are likely simultaneously determined — conflict destroys economic activity, and weak economies may be more conflict-prone. A simple OLS regression of conflict on growth would give biased estimates because of this **endogeneity**.

The solution: Use **rainfall** as an instrument for GDP growth. Rainfall drives agricultural output and economic growth in African economies, but there's no obvious reason it would directly cause civil war except through economic channels.

First, estimate the potentially biased OLS model:

```
ols_model <- lm(any_prio ~ gdp_g + polity2l + ethfrac + Oil + lpopl1,
                data = conflict_data)
summary(ols_model)
```

Now estimate the **IV model** using `ivreg()`. The syntax uses `|` to separate the structural equation from the instruments. Variables that appear on both sides of `|` are exogenous controls; variables that appear *only* on the right side are the excluded instruments.

```
iv_model <- ivreg(any_prio ~ gdp_g + polity2l + ethfrac + Oil + lpopl1 |
                    ... + polity2l + ethfrac + Oil + lpopl1,
                  data = conflict_data)
summary(iv_model, diagnostics = TRUE)
```

> **Hint**
>
> Replace the `...` with the instruments for GDP growth: `GPCP_g + GPCP_g_l` (current and lagged rainfall). The controls (`polity2l`, `ethfrac`, `Oil`, `lpopl1`) appear on both sides because they are assumed to be exogenous.

Compare the two models:

```
stargazer(ols_model, iv_model, type = "text",
          column.labels = c("OLS", "IV"),
          title = "Economic Growth and Conflict: OLS vs IV")
```

> **Reading IV diagnostics**
>
> When you run `summary(iv_model, diagnostics = TRUE)`, look for three things:

> 1. **Weak Instruments F-statistic:** Is $F > 10$? If not, the instruments may be too weak and IV estimates could be unreliable.
> 2. **Wu-Hausman test:** Is $p < 0.05$? If so, endogeneity is present and we *need* IV.
> 3. **Sargan test** (when overidentified): Is $p > 0.05$? If so, the instruments appear valid.

Questions:

- How does the GDP growth coefficient change between OLS and IV?
- Why might OLS be biased here?
- What does the IV estimate tell us about the causal effect of growth on conflict?

Reflection Questions

Write a few sentences for each of the following:

1. **Panel Data:** Why might individual fixed effects be important for studying charitable giving? What unobserved characteristics might bias pooled OLS?

2. **Instrumental Variables:** What two assumptions must hold for rainfall to be a valid instrument for economic growth? Can you think of reasons either assumption might be violated?

3. **Method Choice:** For each analysis in this lab (panel models, mediation, IV), what are the key assumptions? When might these methods fail?